



Joshua Skelton

Technical AI Lead | AI Software Engineer | Applied LLM Systems | Computer Vision

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PROFESSIONAL SUMMARY

AI/software engineer and current Technical AI Lead building applied LLM systems, project-aware AI tools, workflow automation, cloud applications, and computer vision pipelines. Experienced across native C++ AI tooling, FastAPI/Nuxt applications, AWS-oriented portal development, retrieval workflows, model evaluation, and deployment-aware engineering. Strong at translating ambiguous operational requirements into structured, reliable software and reviewable technical delivery.

SELECTED AI / SOFTWARE PROJECTS

- Apr 2026 - **Present** - Selfcaremen telehealth patient portal implementation - Professional patient portal work across secure patient access, lab results, appointments, prescriptions, healthcare workflows, and integration-oriented AWS portal architecture (Nuxt 4, Vue 3, TypeScript, AWS Cognito, API Gateway, Lambda, Elixir, Acuity)
- Nov 2024 - **May 2026** - Coeus AI project-aware desktop workspace - Native C++ AI workspace with custom Skia/SDL2 GUI, source-grounded retrieval, configurable provider workflows, plan/apply tooling, telemetry, headless evaluation reports, and a private technical review package (C++20, Skia, SDL2, OpenGL, retrieval, LLM tooling)
- Dec 2023 - **Dec 2024** - Log and bark computer vision pipeline - End-to-end industrial computer vision pipeline for log detection, stitched log reconstruction, log-boundary segmentation, bark segmentation, and bark-percentage analysis (EfficientNet, PCA, SVM, YOLO segmentation, PyTorch, OpenCV, Python, C++, WSL2)
- Sep 2024 - **Dec 2024** - Agentic digital consultant - Deployed AI consultation workflow using LLM intent classification, deterministic state management, Elai.io-triggered videos, email verification, and business lead handoff on a DigitalOcean Droplet (FastAPI, LangChain, OpenAI, Nuxt, Vue, TypeScript)
- Feb 2023 - **Aug 2023** - Smart-based document chatbot - Retrieval-augmented chatbot combining LLM responses with corporate document data and FAISS-based search (Python, LangChain, FastAPI, FAISS, NGINX)

RELEVANT EXPERIENCE

Technical AI Lead / Software Engineer (Contract) | Selfcaremen - Telehealth Company (Apr 2026 - Present)
Remote / Auckland, NZ

- Leading technical implementation for a professional telehealth patient portal focused on secure patient access, lab results, appointments, prescriptions, and healthcare workflow automation.
- Developing Nuxt/Vue patient-facing frontend flows with Cognito authentication, reusable composables, and integration-ready state management.
- Shaping AWS-aligned portal architecture using Cognito, API Gateway, Lambda-oriented service boundaries, and secure healthcare data-access patterns.
- Working around integration requirements for systems such as Elixir and Acuity while preserving a clear privacy and security boundary for patient data.
- Translating stakeholder and patient-journey requirements into production-oriented portal features and implementation plans.

Machine Learning / Computer Vision Engineer (Contract) | Commercial Client Project (Dec 2023 - Dec 2024)
Auckland, NZ

- Designed a multi-stage computer vision pipeline for commercial log and bark percentage analysis under lightweight Jetson Nano deployment constraints.
- Built and evaluated models across EfficientNet, PCA, SVM, and YOLO segmentation to balance accuracy, runtime cost, and edge-device feasibility.
- Used Python, C++, PyTorch, OpenCV, and WSL2 to support training, validation, inference, and deployment-oriented experimentation.
- Translated a real operational use case into an end-to-end measurable pipeline with emphasis on model selection, reproducibility, and practical deployment trade-offs.

Data Science & AI Tutor | Institute of Data (Nov 2023 - Present)
Auckland, NZ

- Support students and the lead trainer across machine learning, analytics, Python, and project delivery topics.
- Help students debug code, interpret model outputs, and improve the structure of their technical work.
- Provide practical feedback on projects and reinforce core software, data, and machine-learning concepts.
- Collaborate with the lead trainer to support student progress and unblock technical issues.

TARGET ROLES

- AI Software Engineer / Applied AI Engineer
- Forward-Deployed AI Engineer / AI Solutions Engineer
- LLM Applications Engineer / AI Tools Engineer
- Computer Vision Engineer / Full-Stack AI Engineer

EDUCATION

- Auckland University of Technology**
Data Science & Artificial Intelligence
Completed 2023

TECHNICAL SKILLS

- Python, C++, TypeScript, SQL
- FastAPI, Nuxt, Vue, REST APIs, session/state workflows
- LLM applications, LangChain, retrieval workflows, prompt/control design
- Agentic workflows, deterministic state machines, tool-style runtimes
- Forward-deployed engineering: requirements discovery, stakeholder workflows, implementation, delivery
- Computer vision: YOLO segmentation, EfficientNet features, PCA, SVM, PyTorch, OpenCV, scikit-learn
- AWS-aligned architecture: Cognito, API Gateway, Lambda, Amplify-ready frontend design
- Healthcare portal workflows: secure access, lab results, appointments, prescriptions, integration planning
- Deployment and operations: DigitalOcean, environment configuration, structured logging, validation, observability
- Technical documentation, product thinking, tutoring, mentoring, and stakeholder communication

PORTFOLIO EVIDENCE

- Public portfolio with project pages, demo videos, evaluation reports, and technical case studies.
- Private code-review packages available on request for Coeus AI and the digital consultant backend.
- Coeus AI includes a public demo release, RepoGrounding evaluation reports, and clean private codebase metrics.